# ALPHA OS CORE BACKTESTING REPORT

**Asset Class:** XAUUSD (Gold Options / Variance Arbitrage) | **Backtest Window:** April 1, 2026 – April 30, 2026

## Market Regime Context

During the April 2026 macro cycle, spot Gold (XAUUSD) demonstrated a highly volatile profile driven by safe-haven capital flows. The underlying asset opened the month at **$4,699.55**, accelerated violently to mid-month highs of **$4,836.16**, and subsequently retraced to settle within a closing range of **$4,585.85 – $4,638.00**. This environment presented severe structural strains on static implied pricing proxies, providing a rigorous test environment for the boot_opra_real.cpp bare-metal low-latency execution layer.

## Core Performance Report Metrics

| Metric Parameter | Backtest Output Value | Metric Technical Explanation |
|---|---|---|
| Total Signal Triggers | **9 Trades** | Number of execution commands passing hardware filter (max_edge > 500). |
| Directional Accuracy | **66.67%** | Percentage of triggered operations cleanly capturing variance/theta arbitrage. |
| Strategy Total Return | **+9.11%** | Cumulative algorithmic performance across all structural look-ahead blocks. |
| XAUUSD Benchmark Return | **-2.42%** | Simple underlying buy-and-hold spot gold performance across the month. |
| System Generated Alpha | **+8.23%** | Excess structural premium captured independent of spot market directionality. |
| System Generated Beta | **-0.36** | Negative correlation factor validating safe-haven variance shorting behavior. |

## Fixed-Point Statistical Error Matrix

Because the low-latency core strips float operations to run entirely on integer-scaled boundaries (SCALE = 1,000,000), tracking error occurs relative to continuous-time models. Below are the quantitative deviations recorded against true theoretical BSM metrics:

| Error Metric Type | Integer Deviation Value | Risk Mitigation Assessment |
|---|---|---|
| Root Mean Squared Error (RMSE) | **342.16 Units** | Heavy variance observed during mid-month vertical spikes. Driven by absolute mapping of sharp skew expansion. |

| | | |
|---|---|---|
| Mean Absolute Error (MAE) | **276.23 Units** | Linear tracking error reveals systematic under-hedging of outer skews by the fixed-point square-root LUT proxy. |

## Quantitative Desk Takeaways

1. **Structural Edge:** The core engine effectively converted a tracking liability into an execution gate. By setting a conservative hardware signal threshold (max_edge > 500), the system cleanly converted the 276-unit linear tracking bias into a high-probability protective filter.

2. **Gamma Risk Squeeze:** The 33.33% failure rate concentrated heavily around the mid-month rapid spot expansion ($4,836.16). Under vertical delta accelerations, the integer static SQRT_TIME_LUT under-allocates structural variance. To mitigate this risk in multi-month production deployments, we recommend scaling the hardware threshold dynamically based on trailing historical asset regimes.

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*This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.*